The Solar System I

THE SUN AND THE OBJECTS THAT ORBIT IT, INCLUDING THE PLANETS AND SMALLER BODIES, MAKE UP OUR SOLAR SYSTEM.

At the center of the Solar System, the powerful gravitational force of the Sun holds the eight planets in orbit around it.

SEE ALSO

❮ 178–179 Gravity
❮ 226–227 The Earth
❮ 232–233 The Sun
The Solar System II 236–237 ❯

The word "planet" comes from the Greek word "**planetos,**" which means "**wanderer.**"

Scale of the Solar System

Distances in space are so vast that they are hard to imagine. Earth is about 150 million km (93 million miles) from the Sun. To simplify things, astronomers call this distance an astronomical unit (AU)—so Earth is 1 AU from the Sun. Using this scale, Neptune, the furthest planet, is 30 AU from the Sun.

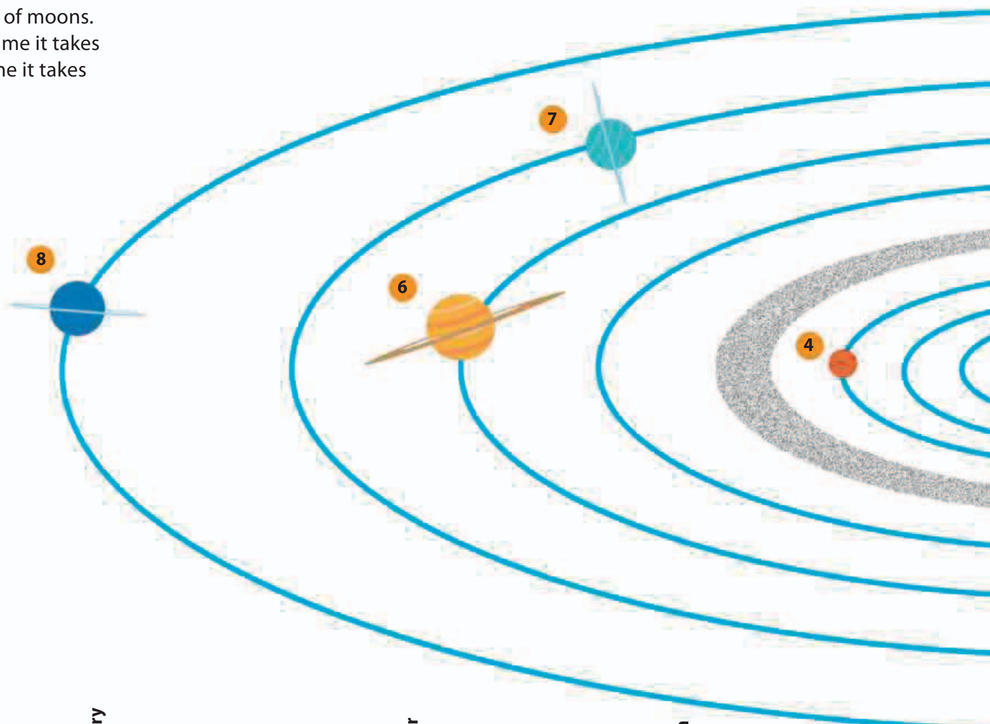
▷ **The planets**
Each planet in the Solar System has its own features, such as distance from the Sun and number of moons. Every planet also has a different year (the time it takes to orbit the Sun), and length of day (the time it takes to rotate once about its axis).

2. Venus
Diameter: 12,104 km (7,521 miles)
Distance from Sun: 0.7 AU
Year: 225 days
Day: 243 days
Number of moons: 0
Average surface temperature: 464°C (867°F)

1. Mercury
Diameter: 4,879 km (3,031 miles)
Distance from Sun: 0.4 AU
Year: 88 days
Day: 58 days
Number of moons: 0
Average surface temperature: 167°C (333°F)

3. Earth
Diameter: 12,756 km (7,926 miles)
Distance from Sun: 1 AU
Year: 365 days
Day: 24 hours
Number of moons: 1
Average surface temperature: 15°C (59°F)

4. Mars
Diameter: 6,786 km (4,217 miles)
Distance from Sun: 1.5 AU
Year: 687 days
Day: 24.5 hours
Number of moons: 2
Average surface temperature: –63°C (–81°F)



▷ **Inner and outer Solar System**
The first four planets—all small and rocky—form the inner Solar System. Beyond the Main Belt, in the outer Solar System, lie the four planets known as the gas giants.